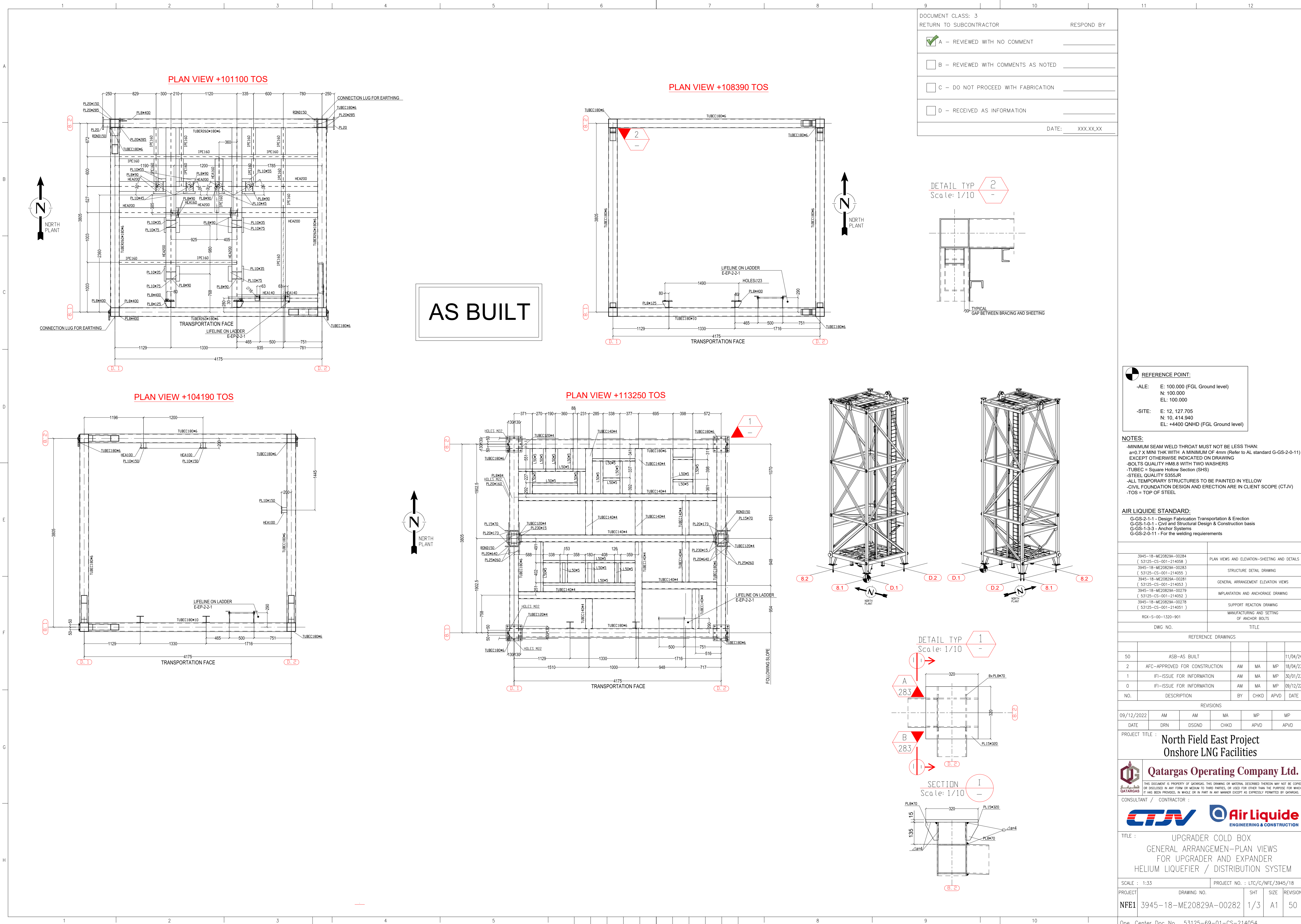




DOCUMENT CLASS: 3	RETURN TO SUBCONTRACTOR	RESPOND BY
☒ A - REVIEWED WITH NO COMMENT		
☐ B - REVIEWED WITH COMMENTS AS NOTED		
☐ C - DO NOT PROCEED WITH FABRICATION		
☐ D - RECEIVED AS INFORMATION		
DATE: XXX.XX.XX		

REFERENCE POINT:	
-ALE:	E: 100.000 (FGL Ground level) N: 100.000 EL: 100.000
-SITE:	E: 12, 127.705 N: 10, 414.940 EL: +4400 QNHD (FGL Ground level)

NOTES:

- MINIMUM SEAM WELD THROAT MUST NOT BE LESS THAN:
a=0.7 X MINI THK WITH A MINIMUM OF 4mm (Refer to AL standard G-GS-2-0-11)
EXCEPT OTHERWISE INDICATED ON DRAWING
- BOLTS QUALITY HM8.8 WITH TWO WASHERS
- TUBEOC = Square Hollow Section (SHS)
- STEEL QUALITY S355JR
- ALL TEMPORARY STRUCTURES TO BE PAINTED IN YELLOW
- CIVIL FOUNDATION DESIGN AND ERECTION ARE IN CLIENT SCOPE (CTJV)
- TOS = TOP OF STEEL

AIR LIQUIDE STANDARD:

- G-GS-2-1-1 - Design Fabrication Transportation & Erection
- G-GS-1-0-1 - Civil and Structural Design & Construction basis
- G-GS-1-3-3 - Anchor Systems
- G-GS-2-0-11 - For the welding requirements

NO.	DESCRIPTION	BY	CHKD	APVD	DATE
3945-18-ME20829A-00284	PLAN VIEWS AND ELEVATION-SHEETING AND DETAILS				
3945-18-ME20829A-00283	STRUCTURE DETAIL DRAWING				
3945-18-ME20829A-00281	GENERAL ARRANGEMENT ELEVATION VIEWS				
3945-18-ME20829A-00279	IMPLANTATION AND ANCHORAGE DRAWING				
3945-18-ME20829A-00278	SUPPORT REACTION DRAWING				
RGX-S-00-1320-901	MANUFACTURING AND SETTING OF ANCHOR BOLTS				
DWG. NO.	TITLE				

NO.	DESCRIPTION	BY	CHKD	APVD	DATE
50	ASB-AS BUILT				11/04/24
2	AFC-APPROVED FOR CONSTRUCTION	AM	MA	MP	18/04/23
1	IFI-ISSUE FOR INFORMATION	AM	MA	MP	30/01/23
0	IFI-ISSUE FOR INFORMATION	AM	MA	MP	09/12/22

NO.	DESCRIPTION	BY	CHKD	APVD	DATE
09/12/2022	AM	MA	MP	MP	
DATE	DRN	DSGND	CHKD	APVD	APVD

PROJECT TITLE : North Field East Project
Onshore LNG Facilities

Qatar Gas Operating Company Ltd.

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CONSULTANT / CONTRACTOR :

Air Liquide
ENGINEERING & CONSTRUCTION

TITLE : UPGRADE COLD BOX
GENERAL ARRANGEMENT-PLAN VIEWS
FOR UPGRADE AND EXPANDER
HELIUM LIQUEFIER / DISTRIBUTION SYSTEM

SCALE : 1:33		PROJECT NO. : LTC/C/NFE/3945/18		
PROJECT	DRAWING NO.	SHT	SIZE	REVISION
NFE1	3945-18-ME20829A-00282	1/3	A1	50

Ope. Center Doc No. 53125-69-01-CS-214054

(ALAIR ref.53125-CS-001-214054 for AIR LIQUIDE purpose only)

Onshore LNG Facilities of the NORTH FIELD EAST PROJECT					
Document Number	3945_18-ME20829A-00282	Document Revision	1	Document Title	UPGRADER COLD BOX - GENERAL ARRANGEMENT PLAN VIEWS FOR UPGRADER AND EXPANDER / HELIUM LIQUEFIER / DISTRIBUTION SYSTEM
Document Receipt Date		Review Code	B	Comments By :	
				Comment Date :	27-Mar-2023

REVIEW CODE DEFINITIONS	
STATUS	REQUIRED ACTION
A- APPROVED / REVIEWED WITH NO COMMENTS	Work associated with the document may proceed - Re-issue without change as next applicable numeric revision
B- APPROVED / REVIEWED WITH MINOR COMMENTS	Revise and issues as next applicable revision. Work Associated with the document may proceed, but comments must
C- RETURNED WITH MAJOR COMMENTS	Revise to address comments and resubmit as next applicable. Work may not proceed until the document is returned
D- NOT ACCEPTED	Alignment meeting between Company and Contractor required to refocus effort. Rework to address comments and
E- INFORMATION ONLY	Receipt Acknowledgement only, any associated work may proceed.
CONTRACTOR'S PERMISSION TO PROCEED OR REVIEW TAKEN ON DOCUMENTS SHALL NOT RELIEVE CONTRACTOR FROM ITS RESPONSIBILITIES OR LIABILITIES	

Comment Number	Discipline	Comment By	Section	CTR Comment	Reply by Air Liquide	Conclusion
1		vsivakumar1	1	COC-Civil Comment: Loads due to Turbine and platform to be considered in Cold box design	Rejected - As agreed during ALE/CTJV meeting, bottom platform shall be linked to the cold box structure in order to have a common independant structure (displacement cold box VS turbine is not allowed) => refer to email "RE: Piping/Civil workshop on upgrader." from O. Riffard dated on 10/02/2023	
2		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
3		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
4		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
5		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
6		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
7		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
8		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
9		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
10		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
11		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
12		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
13		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
14		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
15		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
16		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
17		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
18		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
19		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
20		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
21		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
22		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
23		dcastedo	1	1	All ALE documentation is based on this ALE grid system	
24		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
25		dcastedo	1	A	All ALE documentation is based on this ALE grid system	
26		dcastedo	1	B	All ALE documentation is based on this ALE grid system	
27		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
28		dcastedo	1	2	All ALE documentation is based on this ALE grid system	
29		dcastedo	1	QNHD+12790mm	Equivalence of ALE reference system to QNHD system is already given in drawing	

Onshore LNG Facilities of the NORTH FIELD EAST PROJECT						
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Document Receipt Date		Review Code	B	Comments By :		
				Comment Date :	27-Mar-2023	
30		dcastedo	1	QNHD+5500mm	Equivalence of ALE reference system to QNHD system is already given in drawing	
31		dcastedo	1	FGL+	Equivalence of ALE reference system to QNHD system is already given in drawing	
32		dcastedo	1	FGL+	Equivalence of ALE reference system to QNHD system is already given in drawing	
33		dcastedo	1	FGL+	Equivalence of ALE reference system to QNHD system is already given in drawing	
34		dcastedo	1	FGL+	Equivalence of ALE reference system to QNHD system is already given in drawing	
35		dcastedo	1	QNHD+8590mm	Equivalence of ALE reference system to QNHD system is already given in drawing	
36		dcastedo	1	QNHD+17650mm	Equivalence of ALE reference system to QNHD system is already given in drawing	
37		dcastedo	1	comments made on the previous have not been take in account please update. - GRID tag shall be inline with NFE project - Elevation to be in line with NFE projet, coordinates E100000, N100000,EL100000 have been agreed for ALE 3D model but for all dwg NFE elevation to be shown. please refer to FIVES dwg 00715 is already in QNHD, documents need to be homogenous. - vessel clips for the additionnal pltf and Pipes supports commented are missing please refer to the comment in dwg 281	- Equivalence of ALE reference system to QNHD system is already given in drawing - Elevations are fully compliant with the ALE 3D model - All external structures connected on the cold box (connection plates for platform structures and pipe supports) have to be welded on site by CTJV. Impact on cold box structure has to be checked by ALE based on input data provided by CTJV.	
38		dcastedo	1	10,414.940	Accepted	
39		dcastedo	1	68-Y1201 upgrader have been moved 4m toward north, informed during meeting held in november 2022 and already shown in the 3d share with ALE but no impact for your design.	Accepted	
40		dcastedo	1	origine of equipment its on the bottom of the base plate E 12,127.705 N 10,414.940 EL QNHD+4600	Origin / reference point is based on ALE 3D model: E:100.000 = E:12,127.705 N:100.000 = N:10,414.940 (will be modified based on information from CTJV) EL:+100.000 = EL:+4400 QNHD (FGL ground level)	
41		dcastedo	2	NOT ACCEPTABLE, BRING DOCUMENT IN NFE STD	Equivalence of ALE reference system to QNHD system is already given in drawing	
42		dcastedo	2	NOT ACCEPTABLE, TAG OF GRID TO AS PER COMMENTED	All ALE documentation is based on this ALE grid system	
43		cbiellmann	2	1 : not accepted NFE elevation to be shown, we have agree for the 3d model of ALE to have coordinate E100000, N100000, & EL100000 but in dwg all elevations shall be as per NFE data.	Equivalence of ALE reference system to QNHD system is already given in drawing	
44		dcastedo	2	not accepted - see 1	Equivalence of ALE reference system to QNHD system is already given in drawing	
45		dcastedo	2	not accepted - see 1	Equivalence of ALE reference system to QNHD system is already given in drawing	
46		dcastedo	2	not accepted - see 1	Equivalence of ALE reference system to QNHD system is already given in drawing	